



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CHEMISTRY

9647/03

Preliminary Examination
Paper 3 Free Response Questions

17 August 2011
2 hours

Additional Materials: Data Booklet
Writing Papers

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in the brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together with the cover page provided.

Answer any **four** questions.

- 1** Acrylamide is a compound that is found in many deep fried foods such as French fries and potato chips. Lately, it has been added to the list of substances of very high concern in 2010 as it is believed to be carcinogenic.

- (a)** Acrylamide contains 50.7% by mass of carbon, 7.0% by mass of hydrogen, 19.8% by mass of nitrogen and 22.5% by mass of oxygen. Given that the relative molecular mass of acrylamide is 71.0, determine the molecular formula of acrylamide. Hence, draw a possible structure of acrylamide.

[3]

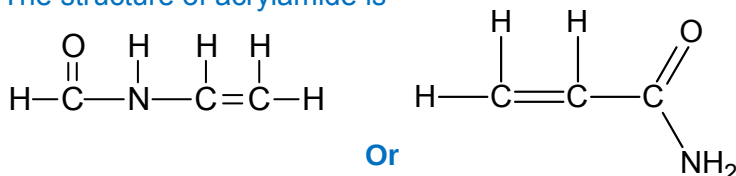
	C	H	N	O
A_r	12	1	14	16
% by mass	50.7	7.0	19.8	22.5
Ratio	4.225	7	1.414	1.406
	3	5	1	1

Let the molecular formula of acrylamide be $C_{3n}H_{5n}N_nO_n$.

$$\begin{aligned} M_r \text{ of acrylamide} &= 71.0 \\ n &= 1 \end{aligned} \quad = n(3 \times 12.0 + 5 \times 1.0 + 14.0 + 16.0)$$

Hence, the molecular formula of acrylamide is C_3H_5NO .

The structure of acrylamide is



- (b)** Apart from deep fried food, acrylamide can also be found in beverages such as prune juice and brewed coffee. Studies have shown that a person can safely ingest up to 0.5 mg / kg body weight in a day without any observed adverse effects to his or her nervous system. Given that 1 kg of brewed coffee contains 1.2×10^{-7} mol of acrylamide, calculate the maximum volume of coffee a 50 kg woman can drink per day. You may assume that coffee has the same density as water.

[3]

$$\begin{aligned} \text{Maximum mass of acrylamide the woman can take in per day} &= 0.5 \times 10^{-3} \times 50 \\ &= 0.025 \text{ g} \end{aligned}$$

$$\begin{aligned} \text{Mass of acrylamide in 1 kg of coffee} &= 1.2 \times 10^{-7} \times 71.0 \\ &= 8.520 \times 10^{-6} \text{ g per kg of coffee} \end{aligned}$$

$$\begin{aligned} \text{Maximum mass of coffee the woman can drink per day} &= 0.025 \div 8.520 \times 10^{-6} \\ &= 2934 \text{ kg} \end{aligned}$$

$$\begin{aligned} \text{Maximum volume of coffee the woman can drink per day} &= 2934 \div 1 \\ &= \underline{\underline{2930 \text{ dm}^3}} \end{aligned}$$

- (c) State the pH of the solution formed when acrylamide dissolves in water. With reference to the structure of acrylamide, explain your answer clearly.

[3]

The pH of the solution is 7

Electron-withdrawing C=O group reduces the electron density on the nitrogen atom and lone pair of electrons on the nitrogen atom is not available to form a dative bond with a proton.

- (d) Acrylamide is produced through the hydrolysis of acrylonitrile. This reaction is catalysed by the enzyme nitrile hydratase.

- (i) In the process of producing acrylamide, the pH and temperature of the system must be carefully maintained so that the enzyme will not undergo denaturation. Explain what is meant by the term *denaturation*.

Denaturation refers to the distortion of shape of proteins by breaking the weak bonds holding the secondary, tertiary and quaternary structure but not the covalent bonds within the primary structure.

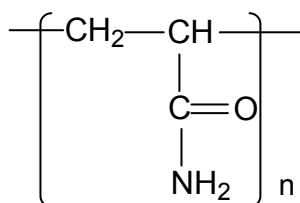
- (ii) Explain in terms of structure and bonding how extreme pH and temperature can cause denaturation of an enzyme.

Extreme pH disrupts the ionic bonds by protonating charged R groups at low pH and deprotonating charged R groups at high pH. This affects the tertiary and quaternary structure of the enzyme, resulting in a more disordered arrangement.

High temperature disrupts the weak Van der Waals' forces holding the quaternary, tertiary and secondary structures, resulting in a more disordered arrangement.

[5]

- (e) When molecules of acrylamide are heated, they undergo polymerisation to form the polymer, polyacrylamide. The structure of polyacrylamide is shown below:



- (i) Polyacrylamide is sometimes used in disposable diapers as it is able to absorb as many times its mass, of water. However, when sodium chloride is added to the polymer, the absorbed water is immediately released. Explain.

The amide group in each subunit of polyacrylamide can form favourable intermolecular hydrogen bonds with water molecules.

Upon addition of ionic sodium chloride, the amide groups can form ion-dipole interactions with the ions which are stronger and consequently more favourable than the intermolecular hydrogen bonds with water.

Or

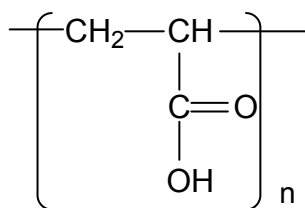
Hence, the intermolecular hydrogen bonds are broken so that ion-dipole interactions can be formed instead.

- (ii) Aqueous solutions of polyacrylamide are found to undergo chemical degradation readily when pH and temperature are high. Identify the type of reaction the polymer has undergone and describe the products formed.

In the presence of OH^- ions and at high temperature, the $-\text{CONH}_2$ side chain of polyacrylamide undergoes hydrolysis.

This will lead to the liberation of NH_3 gas, leaving behind a carboxylate group in the side chain of the polymer.

- (iii) Polyacrylic acid is a polymer that is quite similar to polyacrylamide. The structure of polyacrylic acid is shown below:



Suggest how polyacrylamide can be distinguished from polyacrylic acid, without the use of any strong mineral acids or bases.

Test: Na_2CO_3 (aq), room temperature

Observations: No effervescence with polyacrylamide. Effervescence observed with polyacrylic acid.

Alternative Test: PCl_5 (s)

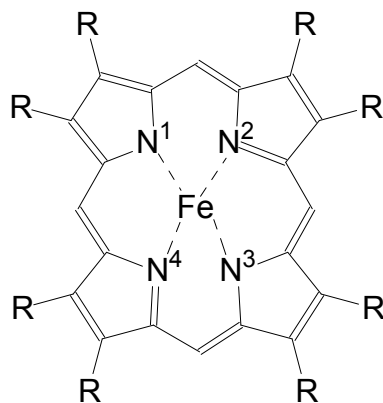
Observations: White fumes observed.

[6]

[Total: 20]

- 2 Metalloproteins is a generic term for a protein that contains a metal ion which is required to assist in the protein's biological activity. Three metalloproteins that can be found in ocean organisms are the iron-based haemoglobin, copper-based haemocyanin and the vanadium-based vanabin.

(a) Haemoglobin has a quaternary structure that contains globular proteins and embedded haem groups. The diagram shows a simplified version of the haem group. The various different side-groups are all shown as R.



Note that the four N atoms and the Fe ion are planar.

(i) Suggest what is meant by the term *ligand*.

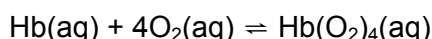
A **ligand** is defined as a neutral molecule or anion which contains at least one **lone pair** of electrons available for forming **dative / coordinate bond** with the **available empty orbitals** on the central metal atom/ion.

(ii) State the difference between the type of bonds that occur between Fe ion and the numbered N¹ and N² atoms.

N¹ – covalent bond

N² – dative bond / coordinate bond

(iii) One molecule of haemoglobin (Hb) can bind up to four molecules of oxygen, according to the following equation.



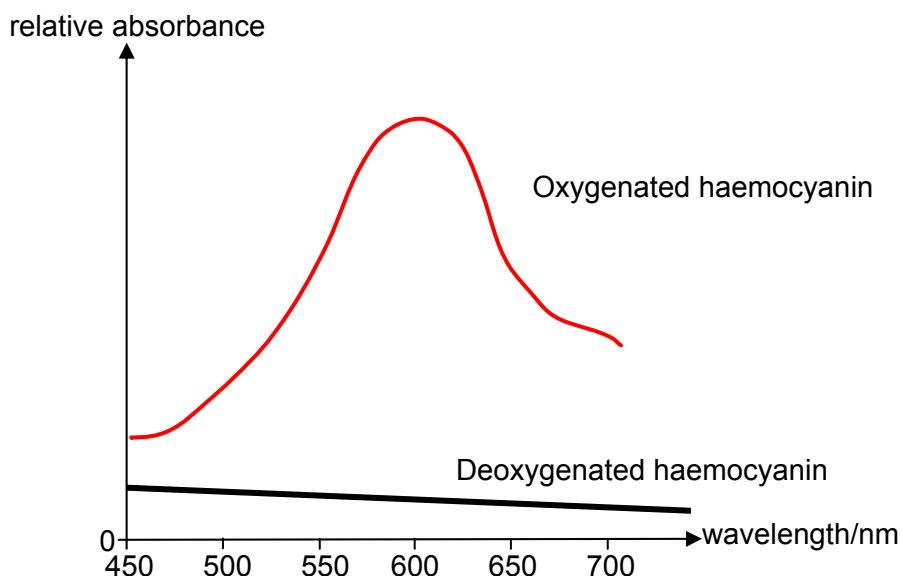
By considering the role of oxygen in this equation, comment on how the body cells receive oxygen from the air in the lungs.

Oxygen acts as a **ligand**.

[O₂] is **higher** in the lungs, **ligand exchange** will occur, resulting in the shifting of the equilibrium position to the **right** to form Hb(O₂)₄ complex.

In the body cells, where the [O₂] is **lower**, equilibrium position will thus shift to the **left**, releasing the O₂ to the body cells.

- (b) The visible absorption spectra of the oxygenated and deoxygenated forms of haemocyanin are shown below:

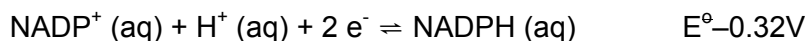


- (i) What are the colours of the two forms of haemocyanin?
 Oxygenated haemocyanin – blue/purple/indigo/violet
 Deoxygenated haemocyanin – colourless/white
- (ii) Explain why the oxygenated haemocyanin is coloured.
 The d orbitals of Cu^{2+} are split into two groups in haemocyanin
 The d electron undergoes d-d transition and is promoted to a higher energy d orbital
 In the process, the d electron absorbs certain wavelength of light energy from the visible region of the electromagnetic spectrum and transmits the remaining wavelength which appears as the colour observed
- (iii) State the full electronic configuration of the copper ion in the deoxygenated haemocyanin.
 $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
- (iv) Suggest the wavelength of the absorption peak for a sample of red-pigmented haemoglobin.
 Student suggest a wavelength between 450 – 550 nm.

[6]

- (c) Vanabin in sea squirts accumulates vanadium in the blood cell by using nicotinamide adenine dinucleotide phosphate (or NADPH) as a reducing agent to give the sea squirts their distinct green colouration.

Use the standard redox potential data in *Data Booklet* and writing equations wherever possible to predict the outcome of the reactions of vanadate (V) ions with excess NADPH. Hence, determine the ion that is responsible for the green colouration in sea squirts.



$$E^\circ = 1.00 - (-0.32) = \underline{+1.32 \text{ V}}$$



$$E^\circ = 0.34 - (-0.32) = \underline{+0.66 \text{ V}}$$



$$E^\circ = -0.26 - (-0.32) = \underline{+0.06 \text{ V}}$$

Since E°_{cell} is $\approx 0 \text{ V}$, **reaction is at equilibrium.**

Hence final reduced vanadium species is V^{2+} or V^{3+} .

[4]

- (d) Group II elements are also being involved in metalloproteins. An example is chlorophyll. Chlorophyll is vital for the process of photosynthesis, and it contains a magnesium ion. Unlike haemoglobin, which utilises the variable oxidation state of iron ion to transport oxygen, chlorophyll makes use of another mechanism to fulfil its function.

- (i) Account for the tendency for transition metals ions to have variable oxidation states.

Transition metals possess variable oxidation states due to the small energy difference between the 3d and 4s electrons.

- (ii) Explain why transition metals carbonates such as copper (II) carbonate often decompose at a lower temperature than Group II carbonates.

Transition metal ions have higher charge and are smaller in size (or higher charge density) compared to Group II metal ions.

Thus, transition metal cations have greater polarising power, hence greater polarizing effect on anion electron cloud.

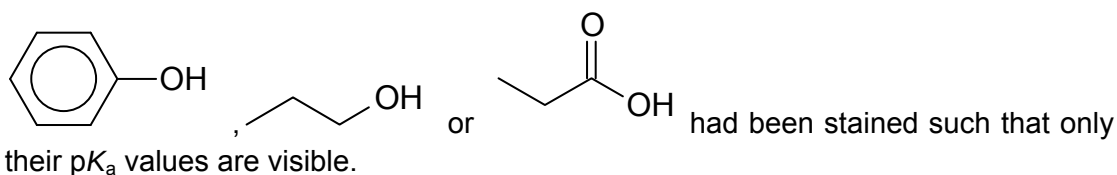
Therefore, less energy are required to decompose transition metal carbonates, thus are less stable to thermal decomposition than Group II metal carbonates.

- (iii) Discuss why the densities of the first transition element series increase along the sequence as far as copper.

Increase in relative atomic mass and gradual decrease in atomic radius [5]

[Total: 20]

- 3 The labels to three bottles of organic compounds each containing either

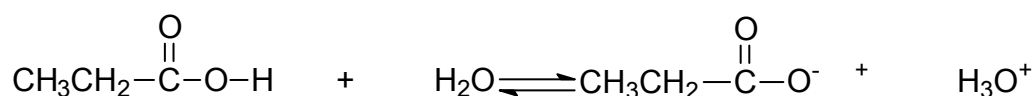


The pK_a values on these 3 bottles read 4.87, 9.95 and 16.0.

- (a) (i) Explain what is meant by *acid dissociation constant*.

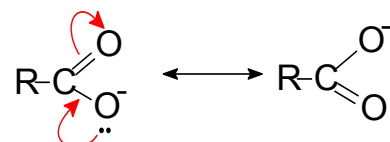
K_a is a measure of the strength of a weak acid.

- (ii) Assign the correct pK_a value to the 3 compounds and explain your answer.

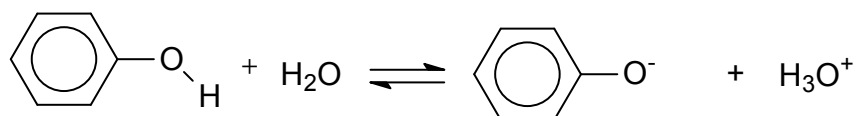


Lone pair of electrons on oxygen interacts with the $-\text{C}=\text{O}$.

OR Carboxylate ions are stabilised by charge delocalisation

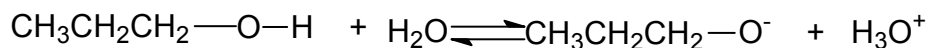


This disperses the negative charge on the 2 oxygen atoms on the carboxylate ion



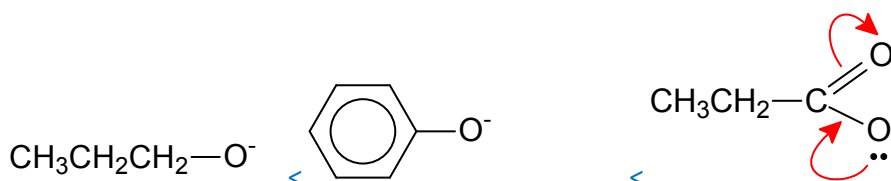
Lone pair of electrons on oxygen interacts with the delocalised π electron cloud of the benzene ring or the phenyl group/benzene ring has a electron withdrawing effect.

This disperses the negative charge on the phenoxide ion.

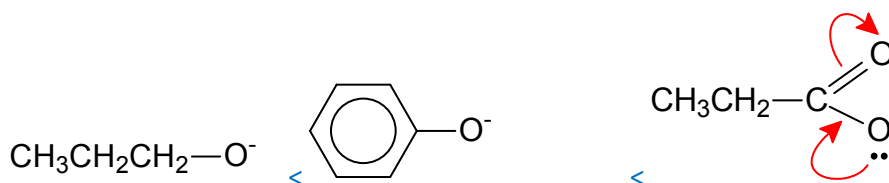


Presence of electron-donating alkyl group intensifies the negative charge on the alkoxide ion

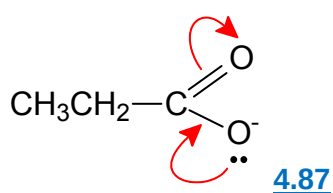
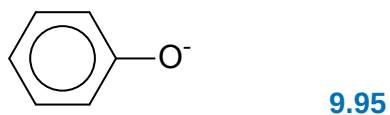
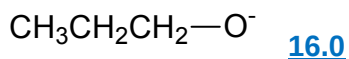
Stability of conjugate base:



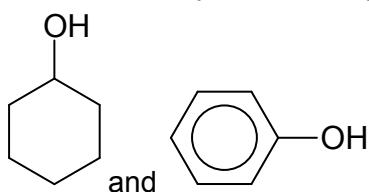
Acid strength:



pKa:



(iii) Describe how you can carry out a simple chemical test to distinguish between



Test: Add neutral FeCl_3 (aq) to both compounds separately.

Observation: Violet complex seen in test tube containing phenol.

No Violet complex seen in test tube containing cyclohexanol.

Or

Alternative Test: Add conc. HNO_3 to both compounds separately.

Observation: White precipitate is formed for phenol.

No white precipitate is formed for cyclohexanol.

Or

Alternative Test: Add PCl_5 to both compounds separately.

Observation: White fumes of HCl is evolved for cyclohexanol.

No White fumes of HCl is evolved for phenol.

Or

Alternative Test: Add dil H_2SO_4 / KMnO_4 , heat to both compounds separately.

Observation: Decolourisation of purple KMnO_4 for cyclohexanol.

No decolourisation of purple KMnO_4 for phenol.

Or

Alternative Test: Add Br_2 (aq) to both compounds separately.

Observation: No decolourisation of brown Br₂(aq) for cyclohexanol.

Brown Br₂(aq) decolourises with formation of white ppt for phenol.

[8]

- (b) Buffered propanoic acid mixes have always been used as preservatives in the process of growing *alfalfa-grass*. It is by far, the most effective in inhibiting growth of moulds. To ensure its effectiveness, the propanoic acid/sodium propanoate buffer system is maintained at pH 6.

- (i) With the aid of equation(s) or otherwise, explain how the buffer system helps to maintain pH.

When a small amount of H⁺ is added,



H⁺ is removed as CH₃CH₂COOH.

When a small amount of OH⁻ is added,



OH⁻ is removed as CH₃CH₂COO⁻ and H₂O.

- (ii) Using the pK_a value of propanoic acid that you have identified in (a)(ii), calculate the ratio of propanoate and propanoic acid in the buffer system.

$$\text{pK}_a = -\lg K_a$$

$$4.87 = -\lg \frac{[\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{CH}_2\text{COOH}]}$$

$$\frac{[\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{CH}_2\text{COOH}]} = 1.35 \times 10^{-5}$$

$$\frac{[\text{CH}_3\text{CH}_2\text{COO}^-]}{[\text{CH}_3\text{CH}_2\text{COOH}]} = \frac{1.35 \times 10^{-5}}{10^{-6}}$$

$$= \underline{13.5}$$

Hence, determine the new pH of a 2 dm³ buffer solution containing 0.05 mol dm⁻³ of propanoic acid when 0.03 mol of H⁺ is added.

$$\text{pH} = \text{pK}_a + \lg \frac{[\text{CH}_3\text{CH}_2\text{COO}^-]}{[\text{CH}_3\text{CH}_2\text{COOH}]}$$

$$[\text{CH}_3\text{CH}_2\text{COO}^-] = 13.5 \times 0.05 = 0.675 \text{ mol dm}^{-3}$$

$$[\text{CH}_3\text{CH}_2\text{COO}^-]_{\text{new}} = [(0.675 \times 2) - 0.03] / 2.0 = 0.66$$

$$[\text{CH}_3\text{CH}_2\text{COOH}]_{\text{new}} = [(0.05 \times 2) + 0.03] / 2.0 = 0.065$$

$$\text{New pH} = 5.88$$

[6]

- (c) An organic compound **A**, $C_5H_8O_2$, gives a weakly acidic solution in water. When hydrogen bromide gas was reacted with **A**, a chiral compound **B**, $C_5H_9O_2Br$ was produced. On heating **B** with aqueous NaOH followed by concentrated H_2SO_4 , a sweet smelling neutral liquid **C** yielded. Compound **C** has the same molecular formula as **A**. When 1 mol of **A** was heated with $KMnO_4$ in dilute H_2SO_4 , 1 mol of an acidic gas was produced.

Deduce the structural formula of **A**, **B**, and **C**.

A has 2 O atoms and gives a weakly acidic solution

→ Presence of COOH

A undergoes electrophilic addition with HBr

→ Presence of alkene/double C-C bond

B undergoes nucleophilic substitution with NaOH and acidification with H_2SO_4

→ The intermediate must contain -OH

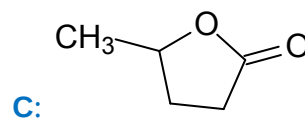
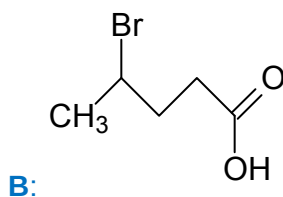
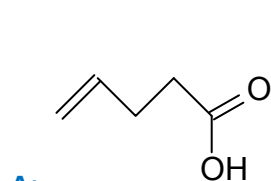
B/intermediate undergo nucleophilic substitution/esterification to form **C**

→ **B** must contain both -OH and -COOH

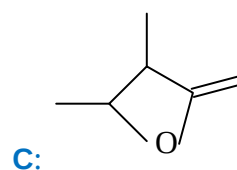
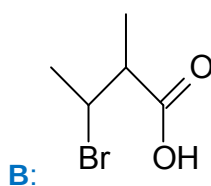
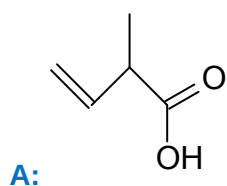
→ **C** must be an ester

1 mol of A undergoes vigorous oxidation with $KMnO_4$ to produce 1 mol CO_2 gas

→ The alkene must be at the terminal



OR



[6]

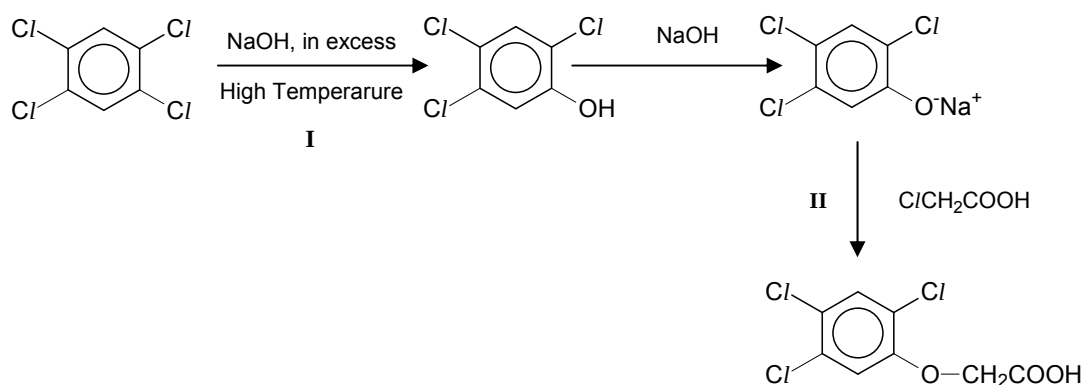
[Total: 20]

4 Herbicides, also known as weedkillers are widely used in agriculture.

2,4,5-Trichlorophenoxyacetic acid (2,4,5-T) was a widely used herbicide until being phased out starting in the late 1970s due to health reasons.

[5]

(a) 2,4,5-T can be synthesised by the following process:

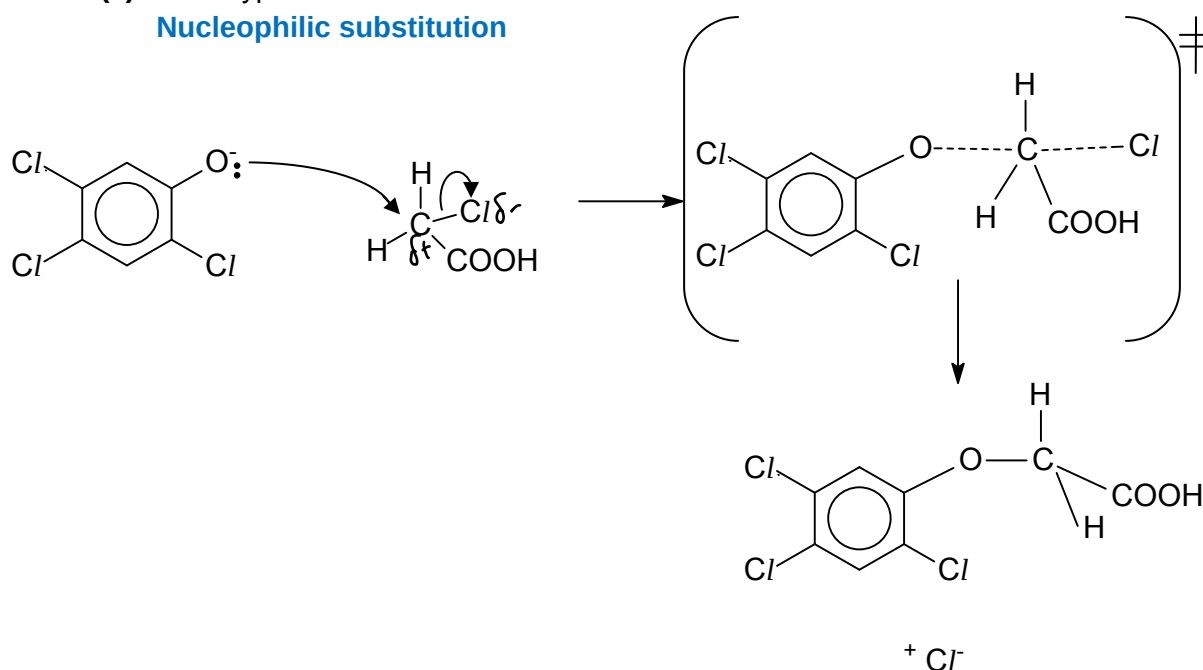


(i) What is unusual about reaction I? Explain your answer.

Chlorobenzene does not usually undergo nucleophilic substitution. This is because the p-orbital of the chloro atom in E interacts/delocalised with the π electron cloud of the benzene ring. This strengthens the carbon-chlorine (C-Cl/ partial double bond) bond.

(ii) What type of reaction is reaction II? Describe the mechanism involved.

Nucleophilic substitution



- (b) To test for presence of halogenoalkanes, silver nitrate is often used to precipitate out any halide ions present after reactions.

An example of such a precipitate is silver chloride. The solubility product of silver chloride at a given temperature is $1.6 \times 10^{-5} \text{ mol}^2 \text{ dm}^{-6}$.

- (i) Calculate the solubility of silver chloride in water.

$$\begin{aligned} [\text{Ag}^+][\text{Cl}^-] &= 1.6 \times 10^{-5} \\ [\text{Ag}^+] &= 4.00 \times 10^{-3} \text{ mol dm}^{-3} \\ \text{Since } \text{Ag}^+ &\equiv \text{AgCl}, \\ [\text{AgCl}] &= \underline{4.00 \times 10^{-3} \text{ mol dm}^{-3}} \end{aligned}$$

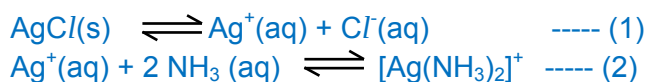
- (ii) Calculate the minimum concentration of dilute hydrochloric acid needed to precipitate out silver chloride if 2 cm^3 of the acid is added to 10 cm^3 of 0.1 mol dm^{-3} of silver nitrate.

$$\begin{aligned} \text{Let the original } [\text{Cl}^-] &\text{ be } a \text{ mol dm}^{-3} \\ \text{New } [\text{Ag}^+] &= 0.1 \times (10/12) = 0.0833 \text{ mol dm}^{-3} \\ \text{New } [\text{Cl}^-] &= a \times (2/12) = 0.1667a \text{ mol dm}^{-3} \\ (0.0833)(0.1667a) &= 1.6 \times 10^{-5} \\ a &= \underline{1.15 \times 10^{-3} \text{ mol dm}^{-3}} \end{aligned}$$

- (iii) Explain what happens when aqueous hydrochloric acid is added to a solution of silver nitrate followed by aqueous ammonia in excess.

Observation:

A white ppt/AgCl (s) is formed and this ppt dissolves when NH_3 is added.

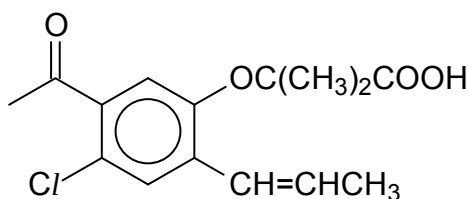


Addition of NH_3 results in the formation of $[\text{Ag}(\text{NH}_3)_2]^+$. This results in the decrease of $[\text{Ag}^+]$. [1]

By Le Chatelier's Principle, the equilibrium position of (1) shifts right.

[6]

- (c) Another common household weed killer, *mecoprop* has the following structure:



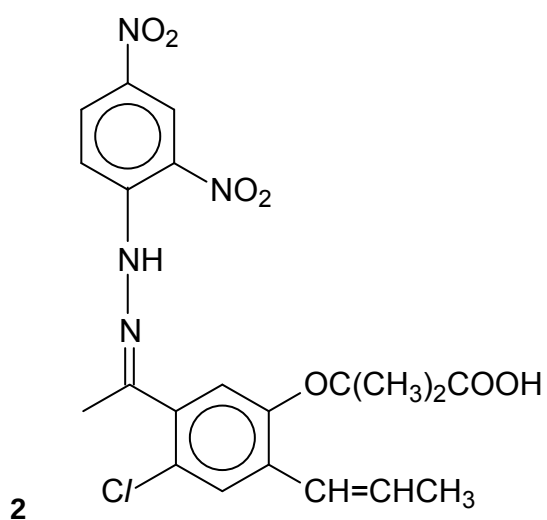
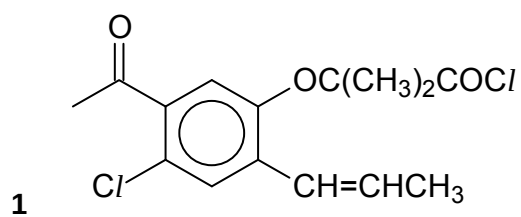
- (i) What type of isomerism is displayed by *mecoprop*?

Geometrical Isomerism

(ii) Draw the structure of the product formed when *mecoprop* reacts with

1 PCl_5

2 2,4- DNPH

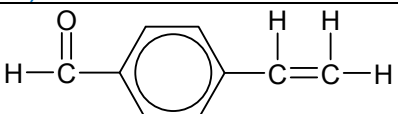
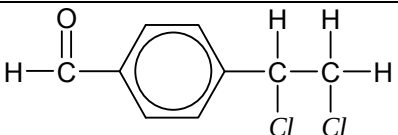
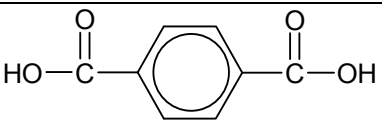


- (d) Compound **D**, C_9H_8O , insoluble in water and aqueous sodium hydroxide, reacts with chlorine gas in the dark to form **E**, $C_9H_8OCl_2$. **D** does not react with Fehling's solution but reacts with Tollens' reagent to give a silver mirror. On heating with acidified potassium manganate(VII), **D** produces **F**, $C_8H_6O_4$, which is soluble in sodium hydroxide.

Deduce the structures of compounds **D**, **E**, and **F**.

[6]

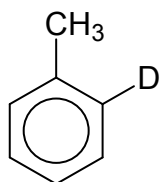
[Total: 20]

d)	Structures: (relative positions of the 2 functional groups on the benzene ring is not important)	
D:		
E:		
F:		
Students need to write both type of reaction and functional group to score the marks.		
	D has <u>comparable no. of C and H atoms with C > 6</u> → D has a <u>benzene ring</u> D does not undergo <u>neutralisation</u> in NaOH → D does not contain <u>phenol</u> and COOH group F undergoes <u>neutralisation</u> with NaOH → F contains <u>COOH group</u> D undergoes <u>electrophilic addition</u> with chlorine gas to form E. → D contains a <u>C=C/double bond</u>, as it D does not undergo <u>oxidation</u> with Fehlings' solution but undergoes oxidation with Tollens' reagent → D contains a <u>benzaldehyde group/aromatic aldehyde</u> D undergoes <u>oxidation</u> to give F → F contains 2 x <u>-COOH</u> groups	

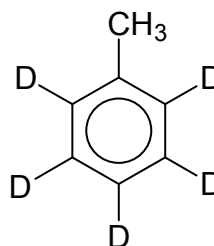
Deuterium (D or ^2H) is a heavy isotope of hydrogen.

Chlorine can react with deuterium to form deuterium chloride, DCl .

- (a) When DCl is dissolved in methylbenzene, no reaction occurs. However, if an anhydrous catalyst **G** is added to the solution, a reaction takes place, and a compound **H** is one of the products. If more DCl is present, a compound **I** is formed.



H



I

- (i) Explain why **G** can act as the catalyst in the reaction.

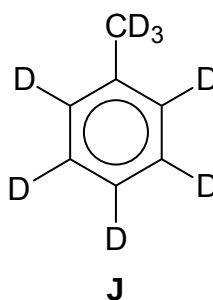
G is able to act as a catalyst for the reaction due to presence of electron-deficient/has available empty d orbital and can accept an electron pair from Cl of the DCl molecule/to generate D^+ .

- (ii) Suggest a suitable identity for the catalyst **G**.

G can be AlCl_3 / FeCl_3 / Fe

- (iii) In the presence of excess DCl , would compound **J** be formed? Explain.

[4]



J

J would not be formed.

The D^+ electrophile would not be attracted to the carbon atom in the $-\text{CH}_3$ group since this carbon atom is not electron-rich. / FRS at the side chain cannot occur in the absence $\text{D}^\bullet$ / free radical.

- (b) Propanone, known industrially as acetone, is an important industrial solvent. It is an active ingredient in nail polish remover.

The bromination of propanone is acid-catalysed:



The rate of disappearance of bromine was measured for several different concentrations of propanone, bromine and hydrogen ions at a certain temperature.

Expt	$[\text{CH}_3\text{COCH}_3] / \text{mol dm}^{-3}$	$[\text{Br}_2] / \text{mol dm}^{-3}$	$[\text{H}^+] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.30	0.05	0.05	5.7×10^{-5}
2	0.30	0.10	0.05	5.7×10^{-5}
3	0.30	0.05	0.10	1.2×10^{-4}
4	0.40	0.05	0.20	3.1×10^{-4}
5	0.40	0.05	0.05	7.6×10^{-5}

- (i) Determine the order of reaction with respect to each reactant and hence the rate equation.

Let the rate equation be $\text{rate} = k[\text{CH}_3\text{COCH}_3]^x[\text{Br}_2]^y[\text{H}^+]^z$

Comparing expt 1 and 5,

$$\left(\frac{0.3}{0.4}\right)^x = \frac{5.7 \times 10^{-5}}{7.6 \times 10^{-5}} \Rightarrow x = 1$$

comparing expt 1 and 2,

$$\left(\frac{0.05}{0.1}\right)^y = 1 \Rightarrow y = 0$$

comparing expt 4 and 5,

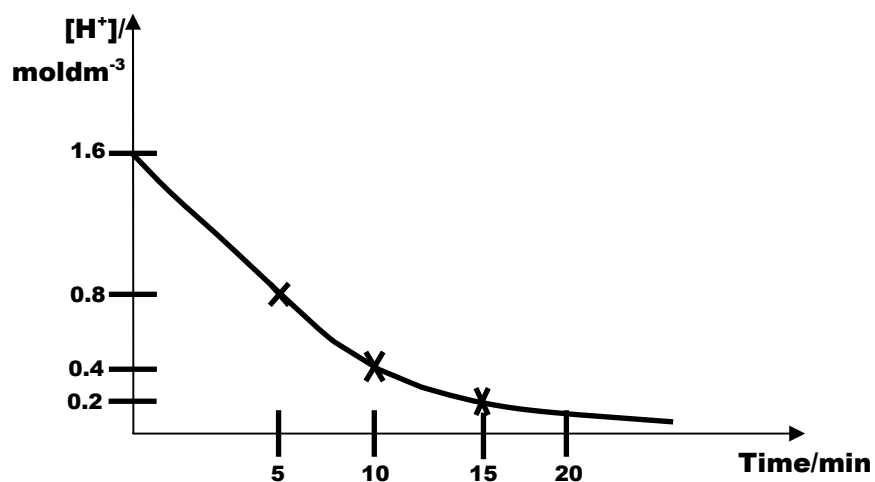
$$\left(\frac{0.2}{0.05}\right)^z = \frac{3.1 \times 10^{-4}}{7.6 \times 10^{-5}} \Rightarrow z = 1$$

Rate = $k[\text{CH}_3\text{COCH}_3][\text{H}^+]$

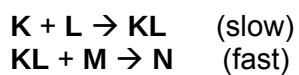
- (ii) In another experiment, the concentration of CH_3COCH_3 used was in excess. As a result, a new rate law was established as follows:

$$\text{Rate} = k' [\text{H}^+]$$

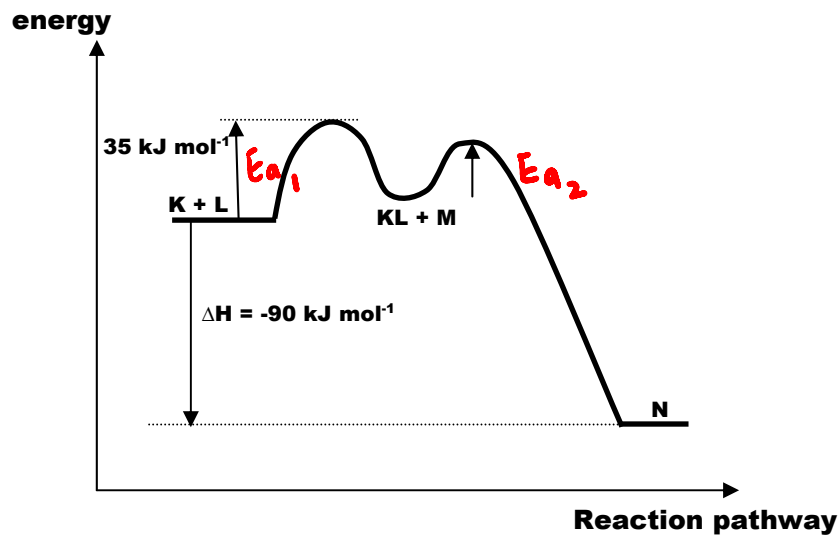
If the initial concentration of H^+ ions is 1.6 mol dm^{-3} and the half-life for the reaction is 5 minutes, sketch a graph of concentration of H^+ against time for the first 20 minutes of the reaction.



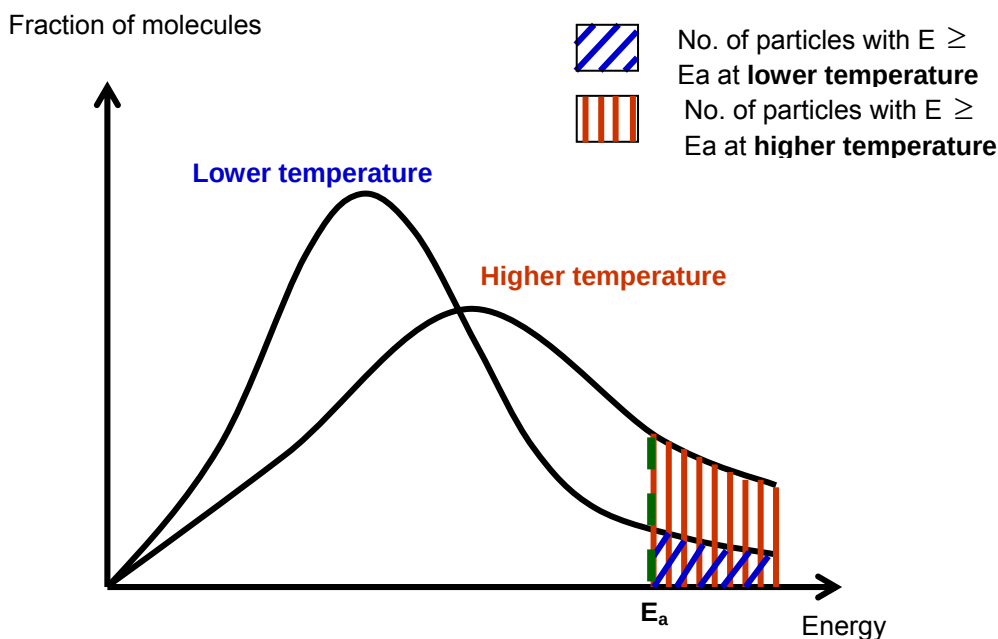
(iii) The following steps in a mechanism are as shown:



Given that the activation energy of the reaction is $+35 \text{ kJ mol}^{-1}$ and ΔH is -90 kJ mol^{-1} , sketch the energy profile diagram for the above reaction.



- (iv) Using an appropriate diagram, explain how higher temperatures affect the rate of reaction.



Diagram

- labeled axes;
- correct shape
- shaded region for higher and lower temperature;
- legend (accept if mentioned in explanation that " area under the graph represents the no. of particles)

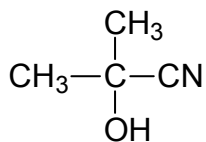
At higher temperature, as depicted by the graph,

- Average kinetic energy of particles $\uparrow$
- No of reactant particles with $E \geq E_a$ $\uparrow$
- No of effective collisions $\uparrow$
- (Since rate directly proportional frequency of effective collisions)
- rate of rxn $\uparrow$

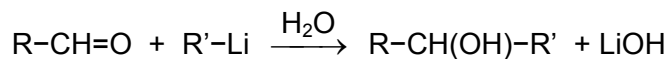
- (v) A quarter of the world's propanone is used to make methyl methacrylate, $\text{CH}_2=\text{C}(\text{CH}_3)\text{CO}_2\text{CH}_3$ which is used on a large scale to produce plastics. The first step is to convert propanone to its corresponding cyanohydrin. The cyanohydrin is then further reacted to form methyl methacrylate.
- State the reagents and conditions required to convert propanone to its corresponding cyanohydrin. Draw the structure of the cyanohydrin.

[13]

HCN with trace KCN OR HCN with NaOH(aq), 10 – 20 °C/cold

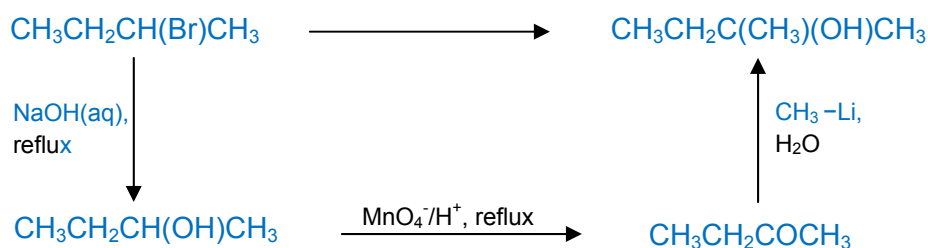


- (c) Organometallic compounds, usually a metal attached to an R group, can be used to convert carbonyl compounds to alcohols. An example is shown below:



2-bromobutane can be converted to 2-methylbutan-2-ol in **3 steps**. Write the reagents and conditions required for the conversion, including intermediate compounds.

[3]



1st step: aq NaOH, reflux and $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

2nd step: dil or aq $\text{H}_2\text{SO}_4/\text{KMnO}_4$, reflux and $\text{CH}_3\text{CH}_2\text{COCH}_3$

3rd step: CH_3-Li (and H_2O)

[Total: 20]

-----End of Paper-----